Software Requirement Specification

Functional Requirements

1.The system shall prompt the user to scan a QR code to provide Hexadecimal values to the system.
2. Each Hexadecimal character shall either be 1 or 2 digits in length.
3. Valid Hexadecimal characters are: 0, 1, 2, 3, 4,5, 6, 7, 8, 9, A, B, C, D, E, F.
4. Hexadecimal letters (A, B, C, D, E, F) shall be case insensitive meaning both upper and lowercases shall be accepted.
5. The system shall allow a maximum of five hexadecimal values per QR code.
6. The system shall accept one or more Hexadecimal values in a single QR code.
7. When multiple Hexadecimal values are provided in a single QR code, they shall be separated by the ‘&’ character.
8. The system shall validate each Hexadecimal character consists of 1 or 2 digits using only characters 0-9 and A-F (case insensitive).
9. the system shall reject any Hexadecimal values that do not meet the criteria of validity.
10. The system shall ignore any invalid hexadecimal values.
711. The system shall notify the user of any hexadecimal values that were ignored due to invalid formatting.
12. The system shall convert each valid hexadecimal value into its decimal equivalent without using java built in conversion methods.
13. The system shall convert every valid hexadecimal value into its binary equivalent without using javas built-in conversion methods.
14. The system shall convert each Hexadecimal value into its octal equivalent without using Java’s built-in conversion methods.
15. The system shall calculate the Wheel speed of the Swiftbot, using the equivalent values of octal to Hexadecimal value.
16. If the octal value is < 50, then the system shall set the speed to octal value +50.
17. if the octal Value is >=50, then the system shall set speed to the given value.
18. If the system detects the calculation exceeds the Swiftbot’s max wheel spin, the system shall set the speed to the maximum wheel spin possible.
19. The system shall calculate the RED LED colour as the decimal equivalent of the Hexadecimal value.
20. The system shall calculate the GREEN LED colour as the remainder of the decimal when divided by 80 then multiplied by 3.
21. The system shall calculate the BLUE LED colour as the greater value between the RED and the GREEN components
22. The system shall determine the sequence of movements of the Swiftbot, which are determined by the binary equivalent.
23. The system shall read the binary number from RIGHT to LEFT.
24. The Swiftbot shall move forward if the binary digit is currently at 1
25. The Swiftbot shall spin if the binary digit is currently at 0
26. If the hexadecimal value is 1 digit in length, each forward movement shall last 1 second.
27. If the hexadecimal value is 2 digits in length, each forward movement shall last 0.5 seconds.
28. The system shall execute all movements in the order determined by the binary representation from right to left.
29. When multiple Hexadecimal values are given, the system shall execute their dance routines in the order they appear in the QR code.
830. Before the Swiftbot begins moving, the system shall display the Hexadecimal value.
31. Before the Swiftbot begins moving, the system shall display the Octal equivalent.
32. Before the Swiftbot begins moving, the system shall display the Decimal equivalent.
33. Before the Swiftbot begins moving, the system shall display the binary equivalent.
34. Before the Swiftbot begins to move, the system shall display the calculated wheel speed.

35. Before the Swiftbot begins to move, the system shall display the RGB (RED, GREEN, BLUE) LED colour values.
36. After completing all movements in the Dance routine, the system shall turn off the Swiftbots under lights.
37. After the under lights turn off, the system shall prompt the suer to either scan another QR code or terminate the program.
38. To scan another QR code, the user shall press ‘Y’ button on the Swiftbot.
39. When the ‘Y’ button is pressed the system shall prompt the user to scan another QR Code.
40. To terminate the program, the user shall press button ‘X’ on the Swiftbot.
41. The system shall store all valid hexadecimal values entered during execution.
42. The system shall sort the stored hexadecimal values when the user has entered so far in ascending order
43. The system shall write the sorted hexadecimal values to a text file upon program termination.
44. The system shall display the file location of the log file to the user before exiting.

Non – Functional Requirements:

45. The system shall provide clear instructions and messages to guide the user through the program.
46. The system shall display informative messages when invalid input is detected.
47. The system shall continue execution when invalid hexadecimal values are provided, rather than terminating unexpectedly.
48. The system shall ensure that all valid user inputs are processed correctly.
949. The system shall complete Hexadecimal validation and conversion for each value within 100 milliseconds